

❖ **Branching Processes**

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Motivation

“How many male children (on average) must each generation of a family have in order for the family name to continue in perpetuity?”

-Sir Francis Galton, 1873

An answer came from Henry William Watson soon after in 1874.

Bienaymé had previously published the answer to the problem in 1845.

A Branching process is a discrete-time Markov process that models a population in which each individual in generation n produces a random number of offsprings for generation $n + 1$. It is a gateway to:

- extinction vs survival,
- phase transitions,
- martingales,
- random trees,
- epidemics, genetics, queueing, and networks.

Generation n

$X_n :=$ Size of population in n -th generation

$n = 0$

$$X_0 = 1$$

$n = 1$

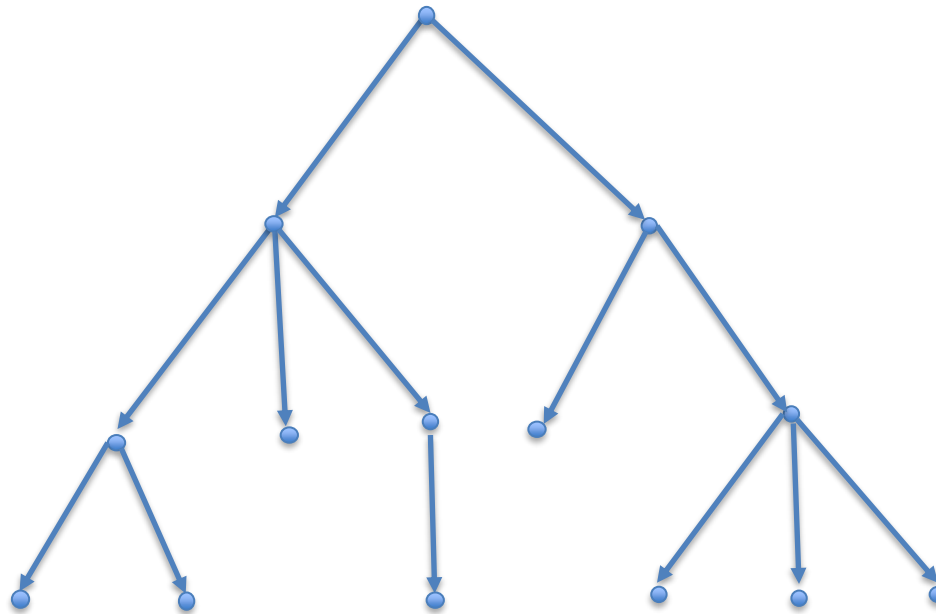
$$X_1 = 2$$

$n = 2$

$$X_2 = 5$$

$n = 3$

$$X_3 = 6$$



$Z_{n,i} :=$ Number of offspring born by the i -person in the $n - 1$ generation.

$$X_n = \sum_{i=1}^{X_{n-1}} Z_{n,i}$$

For example,

$$X_3 = Z_{3,1} + Z_{3,2} + Z_{3,3} + Z_{3,4} + Z_{3,5} = 2 + 0 + 1 + 0 + 3 = 6$$

Example. Z is uniform on $\{0,1,2,3\}$

$$P(Z = k) = \frac{1}{4}$$

Example. Z is Geometric on $\{0,1,2,3, \dots\}$

$$P(Z = k) = p^k(1 - p)$$

Galton–Watson Branching Process

Let

X_n : Population size in generation n

$X_0 \in \mathbb{N}$ given initial population size.

Each individual produces Z offspring according to:

$$\mathbb{P}(Z = k) = p_k \text{ for } k = 0, 1, 2, \dots$$

Assume

- $Z_{n,i}$ are i.i.d. copies of Z
- Independent across generations and individuals

Then, the branching process model is defined by the random sum:

$$X_n = \sum_{i=1}^{X_{n-1}} Z_{n,i}$$

Assume $p_0 \neq 0$. Otherwise, zero is not a state.

Zero state is absorbing, so it is recurrent.

Lemma. In a branching process, all non-zero states are transient.

Total return probability: $f_{ii} = P(\text{return to } i | X_0 = i)$

$$= P(X_n = i \text{ for some } n \geq 1 | X_0 = i)$$

$$\leq P(X_1 > 0 | X_0 = i)$$

$$= 1 - P(Z_1 = 0 | X_0 = i)$$

$$= 1 - p_0^i$$

$$< 1$$

Expected population size

Let m be the expected population after each individual

$$\mu := E[Z] = \sum_{k=0}^{\infty} k p_k$$

Then,

$$E[X_n | X_{n-1} = x_{n-1}] = \sum_{i=1}^{x_{n-1}} E[Z_{n,i}] = \mu x_{n-1}$$

$$E[X_n | X_{n-1}] = \mu X_{n-1}$$

By the law of total expectation, we have

$$E[X_n] = E[E[X_n | X_{n-1}]] = E[\mu X_{n-1}] = \mu E[X_{n-1}]$$

$$\text{So, } E[X_n] = \mu^n E[X_0] = \mu^n \quad \text{Since } X_0 = 1$$

Classification by mean offspring number

$$\lim_{n \rightarrow \infty} E[X_n] = \lim_{n \rightarrow \infty} \mu^n = \begin{cases} 0 & \text{if } \mu < 1 \\ 1 & \text{if } \mu = 1 \\ \infty & \text{if } \mu > 1 \end{cases}$$

Regime	Condition	Mean behavior
Subcritical	$\mu < 1$	Mean decays
Critical	$\mu = 1$	Mean constant
Supercritical	$\mu > 1$	Mean grows

Growing expectation does NOT guarantee survival.

Variance

Let $\sigma^2 = \text{Var}(Z)$ be the variance of Z .

By the law of total variance

$$\begin{aligned}\text{Var}(X_n) &= E[\text{Var}(X_n|X_{n-1})] + \text{Var}(E[X_n|X_{n-1}]) \\&= E[X_{n-1}]\sigma^2 + \mu^2\text{Var}(X_{n-1}) \\&= \mu^{n-1}\sigma^2 + \mu^2\text{Var}(X_{n-1}) \\&= \mu^{n-1}\sigma^2 + \mu^2(\mu^{n-1}\sigma^2 + \mu^2\text{Var}(X_{n-2})) \\&= \dots \\&= \sigma^2\mu^{n-1}(1 + \mu + \mu^2 + \dots + \mu^{n-1}) + \mu^{2n}\text{Var}(X_0) \\&= \sigma^2\mu^{n-1}(1 + \mu + \mu^2 + \dots + \mu^{n-1})\end{aligned}$$

□ Extinction Probability

The probability of extinction is the probability that X_n is eventually zero.

Define **probability of extinction**

$$\rho = \mathbb{P}(X_n = 0 \text{ eventually})$$

Since $\{X_n = 0\} \Rightarrow \{X_{n+1} = 0\}$, we have

$$\{X_n = 0\} \subseteq \{X_{n+1} = 0\}$$

So,

$$(X_n = 0 \text{ eventually}) = \bigcup_{n \geq 0} \bigcap_{k \geq n} \{X_k = 0\}$$

$$\rho = \lim_{n \rightarrow \infty} \mathbb{P}(X_n = 0)$$

Case 1. $\mu < 1$

Lemma. If $\mu = E[Z] < 1$, then $\rho = \mathbb{P}(X_n = 0 \text{ eventually}) = 1$.

Proof.

$$\begin{aligned} E[X_n] &= \sum_{k \geq 1} k \mathbb{P}(X_n = k) \\ &\geq \sum_{k \geq 1} \mathbb{P}(X_n = k) \\ &= \mathbb{P}(X_n > 0) \end{aligned}$$

So,
$$\rho = \lim_{n \rightarrow \infty} \mathbb{P}(X_n = 0) = \lim_{n \rightarrow \infty} 1 - \mathbb{P}(X_n > 0) \geq \lim_{n \rightarrow \infty} (1 - \mu^n) = 1$$

So $\rho = 1$

Probability Generating Functions (PGF)

For a discrete random variable X taking values in $\{0,1,2, \dots\}$, the **Probability Generating Function** of X is the function

$$G_X(x) := E(s^X) = \sum_{k=0}^{\infty} s^k \mathbb{P}(X = k)$$

The function is a power series whose coefficients are probabilities.

$$G_X(0) = P(X = 0)$$

$$G_X'(0) = P(X = 1)$$

$$G_X(1) = 1$$

$$G_X'(1) = E(X)$$

$$\frac{G_X^{(j)}(0)}{j!} = P(X = j)$$

$$Var(X) = G''(1) + G'(1) - G'(1)^2$$

Example. Z is uniform on $\{0,1,2,3\}$

The PGF of Z is

$$G(s) = E[s^Z] = \frac{1}{4} + s \left(\frac{1}{4} \right) + s^2 \left(\frac{1}{4} \right) + s^3 \left(\frac{1}{4} \right) = \frac{1}{4} (1 + s + s^2 + s^3)$$

Example Z is Geometric on $\{0,1,2,3, \dots\}$

The PGF of Z is

$$\begin{aligned} G(s) = E[s^Z] &= \sum_{k=0}^{\infty} s^k p(1-p)^{k-1} = sp \sum_{k=0}^{\infty} (s(1-p))^{k-1} \\ &= \frac{sp}{1 - s(1-p)} \end{aligned}$$

Lemma1. If X and Y are random variables such that $G_X(s) = G_Y(s)$ for all s , then X and Y have the same distribution.

Lemma2. Assume that X and Y are independent,

$$G_{X+Y}(s) = G_X(s)G_Y(s)$$

Lemma3. Suppose N and $Z_1, \dots, Z_N$ are independent and Z_i are iid.

Let $X = Z_1 + \dots + Z_N$.

Then, $G_X(s) = G_N(G_Z(s))$

Proof

Lemma1: by the uniqueness of Power series expansion.

Lemma2: by the independency and property of expected value of products.

Lemma3: By law of total expectation

$$\begin{aligned} G_X(s) &= E[s^X] = E[E[s^X | N]] \\ &= \sum_n E[s^X | N = n] \mathbb{P}(N = n) \\ &= \sum_n E[s^{Z_1 + \dots + Z_N} | N = n] \mathbb{P}(N = n) \\ &= \sum_n E[s^{Z_1}] \dots E[s^{Z_n}] \mathbb{P}(N = n) \\ &= \sum_n (G_X(s))^n \mathbb{P}(N = n) = G_N(G_X(s)) \end{aligned}$$

Some Important Probability Generating Functions

Distribution	PMF / pdf	Support	PGF ($G_X(s) = \mathbb{E}[s^X]$)
Bernoulli (p)	$p^x q^{1-x}$	$\{0,1\}$	$q + ps$
Binomial (n, p)	$\binom{n}{x} p^x q^{n-x}$	$\{0,1, \dots, n\}$	$(q + ps)^n$
Poisson (μ)	$\frac{e^{-\mu} \mu^x}{x!}$	$\{0,1,2, \dots\}$	$e^{-\mu(1-s)}$
Geometric (p)	$q^{x-1} p$	$\{1,2, \dots\}$	$\frac{ps}{1 - qs}$
Negative Binomial (r, p)	$\binom{x-1}{r-1} q^{x-r} p^r$	$\{r, r + 1, \dots\}$	$\left(\frac{ps}{1 - qs} \right)^r$

Here $q = 1 - p$

Case 2. $\mu = 1$ or $\mu > 1$

The PGF of the offspring variable Z is $f(s) := G_Z(s) = E[s^Z]$

The PGF of $X_n = \sum_{i=1}^{X_{n-1}} Z_{n,i}$ is $f_n(s) := G_{X_n}(s) = E[s^{X_n}]$

Lemma. Assume $X_0 = 1$

1. $f_{n+1}(s) = f_n(f(s)) = \dots = f^{n+1}(s) = f(f_n(s))$
2. $f_n(0) = \mathbb{P}(X_n = 0)$ is increasing on n , since $\{X_n = 0\} \subset \{X_{n+1} = 0\}$

$$r := \lim_{n \rightarrow \infty} f_n(0)$$

3. $r = f(r)$

4. $f(0) = \mathbb{P}(Z = 0)$

$$f(1) = 1$$

$$f'(1) = \mu$$

5. $f'(s) = E[Zs^{Z-1}] > 0$

So, $f(s)$ is increasing.

6. $f''(s) = E[Z(Z-1)s^{Z-2}] \geq 0$ if $s \geq 0$

So, $f(s)$ is convex.

Extinction Probability

Theorem:

The probability of extinction r is the **smallest solution in $[0, 1]$** of the fixed-point equation of the PGF of Z :

$$r = f(r)$$

Moreover,

$$r = \begin{cases} 1 & \text{if } \mu < 1 \\ \text{smallest solution of } r = f(r) & \text{if } \mu > 1 \\ 1 & \text{if } \mu = 1 \text{ and } \sigma^2 \neq 0 \\ 0 & \text{if } \mu = 1 \text{ and } \sigma^2 = 0 \end{cases}$$

Hence,

- If $\mu \leq 1$, then $r = 1$ (extinction occurs almost surely)
- If $\mu > 1$, then $0 < r < 1$ (survival with positive probability)

Example:

Example:

❖ References:

Recommended textbooks

- Introduction to Probability Models - Sheldon M. Ross
- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
- Introduction to Stochastic Processes with R - Robert Dobrow

More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
- Markov Chains –J.R. Norris
- Stochastic Processes, an Introduction. –Peter Jones and Peter Smith
- Stochastic Processes 2nd Edition by Sheldon M. Ross